

AS Chemistry homework · 3.2 Ionic bonding

CIE 9701 AS Chemistry · 17 marks · about 40 minutes

Topic: 3.2 Ionic bonding — electrostatic attraction between oppositely charged ions, and describing ionic bonding in NaCl, MgO and related giant ionic compounds. Answer in English. Section A is Paper 1 style (1 mark each). Section B is written; use CIE language (electrostatic attraction, cations/anions, giant ionic lattice, mobile ions when molten).

Section	Content	Marks
A	Multiple choice	7
B1	2022 ON Paper 22 Question 1(a)	2
B2	2021 MJ Paper 22 Question 2(a)	3
B3	2025 MJ Paper 21 Question 1(a)(i) and (iii)	2
B4	2022 ON Paper 23 Question 3(a)(i)	1
B5	2025 MJ Paper 23 Question 2(c)(ii)	2
Total		17

Section A — Multiple-choice questions (7 marks)**A1.**

[1]

Which type of bonding is **never** found in elements?**A** covalent**B** ionic**C** metallic**D** van der Waals' forces

2021 MJ Paper 13 Question 6

A2.

[1]

The atomic radii and ionic radii for three elements in Period 3 are shown.

	atomic radius / nm	ionic radius / nm
element X	0.118	0.053
element Y	0.099	0.180
element Z	0.160	0.072

Using this data, which statement is correct?

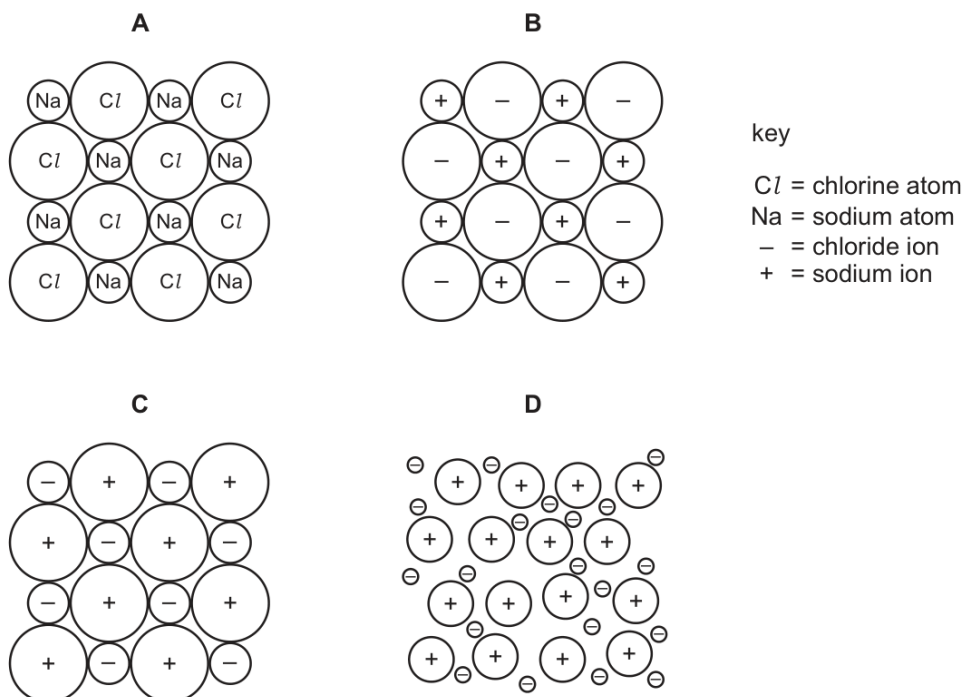
A Element X has lower electrical conductivity than element Y.**B** Element Y has a higher melting point than element X.**C** Element Y and element Z react to form an ionic compound.**D** Element Z forms ionic compounds by gaining electrons.

2025 ON Paper 11 Question 16

A3.

[1]

Which diagram represents the lattice structure of sodium chloride?



2025 MJ Paper 11 Question 8

A4.

[1]

In the sodium chloride lattice the number of chloride ions that surround each sodium ion is called the coordination number of the sodium ions.

What are the coordination numbers of the sodium ions and the chloride ions in the sodium chloride lattice?

	coordination number of sodium ions	coordination number of chloride ions
A	4	6
B	6	4
C	6	6
D	8	6

2023 MJ Paper 11 Question 6

A5.

[1]

Which oxide has a simple structure rather than a giant structure?

A MgO **B** Al₂O₃ **C** SiO₂ **D** P₄O₁₀

2025 ON Paper 13 Question 18

A6.

[1]

Which solid compound has both ionic and covalent bonding but **not** coordinate bonding?

A Al₂Cl₆ **B** CH₃COONa **C** MgCl₂ **D** NH₄Cl

2025 MJ Paper 14 Question 12

A7.

[1]

The following statements compare some properties of the oxides and chlorides of sodium and silicon.

Which statements are correct?

1. The melting point of SiCl₄ is higher than the melting point of SiO₂.
2. SiCl₄ reacts with water, NaCl does not.
3. The melting point of SiCl₄ is lower than the melting point of NaCl.

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

2021 ON Paper 12 Question 35

Section B — Structured questions (10 marks)

B1 · 2022 October/November Paper 22 Question 1(a) (2 marks)

Species such as NH₄⁺, CO₃²⁻ and PO₄³⁻ are examples of molecular ions.

B1(a)

[2]

Ionic and covalent bonds both involve an electrostatic attraction between different species.

Identify the species that are electrostatically attracted to one another in:

- an ionic bond

- a covalent bond.

B2 · 2021 May/June Paper 22 Question 2(a) (3 marks)

The strength of interaction between particles determines whether the substance is a solid, liquid or gas at room temperature. Lithium sulfide, Li₂S, is a crystalline solid with a melting point of 938 °C. It conducts electricity when it is molten.

B2(a)(i)

[1]

Give the formulae of the particles present in solid lithium sulfide.

B2(a)(ii)

[2]

Explain, in terms of the structure of the crystalline solid, why lithium sulfide has a high melting point.

B3 · 2025 May/June Paper 21 Question 1(a)(i) and (iii) (2 marks)

Solid sodium conducts electricity. Sodium oxide conducts electricity when molten but not when solid.

B3(a)(i)

[1]

Name the type of bonding present in sodium and in sodium oxide.

bonding in sodium

bonding in sodium oxide

B3(a)(iii)

[1]

Explain why sodium oxide conducts electricity when molten.

B4 · 2022 October/November Paper 23 Question 3(a)(i) (1 mark)

Some of the common chlorides of Period 3 elements are shown in the list.

NaCl MgCl₂ AlCl₃ SiCl₄ PCl₅

B4(a)(i)

[1]

From this list, identify all the chlorides that have giant ionic structures in the solid state.

B5 · 2025 May/June Paper 23 Question 2(c)(ii) (2 marks)

The chemical properties of oxides are related to the chemical bonding present in these compounds.

Table 2.1 shows the melting points of some oxides.

oxide	melting point / °C
SO ₂	−73
H ₂ O	0
SO ₃	17
SiO ₂	1610
Na ₂ O	1132
MgO	2852
Al ₂ O ₃	2072

Table 2.1

B5(c)(ii)

[2]

A student suggests the following hypothesis.

The melting point of an ionically bonded oxide is only determined by the charge on the cation.

Use Table 2.1 to deduce if this hypothesis is true or false or if there is **not** enough information to make a conclusion. Explain your answer.
